

PUBLICATIONS &
CONFERENCES

[1] Goswami, M., Paul, A. & **Goswami, Meghraj**. (2026). *Performance Comparison of Linear Relation Based and Machine Learning Based Rainfall-Runoff Models for Flow-Simulation for a Data-Scarce River Valley Project*. In: Nguyen, G-N., Swaroop, A., Shukla, P. (Eds.) *Proceedings of Fifth International Conference on Computing and Communication Networks (ICCCN 2025)*. Lecture Notes in Networks and Systems, vol 1773, pp. 56-70. Springer Nature Switzerland AG.
DOI: [10.1007/978-3-032-14197-2_6](https://doi.org/10.1007/978-3-032-14197-2_6)
Presented at ICCNet-2025 held in Manchester Metropolitan University, Manchester, United Kingdom during 01-03 August 2025.

[2] Goswami, M., Rashid, U., Patel, R. & **Goswami, Meghraj**. (2025). *Impacts of Building Rehabilitation by Horizontal and Vertical Extensions of Framed Structures: Risks and Realities*. Presented at the 2nd International Conference on Condition Assessment Rehabilitation & Retrofitting of Structures (CARRS 2025) at Indian Institute of Technology Roorkee, India during 11-13 December 2025 (accepted for publication in Scopus-indexed *Lecture Notes in Civil Engineering* by Springer).

[3] Goswami, M., Paul, A. & **Goswami, Meghraj**. (2024). *Future Precipitation Characteristics of eight Tier I Urban Conglomerates amongst Designated Smart Cities of India under selected Shared Socio-economic Pathways*. Submitted in *Journal of Environmental Processes* by Springer Nature, currently under 'editorial assignment'. Presented at the International Conference on Sustainable Urban Water Management in Fast Growing Cities (SUWM2024) held in BITS Pilani, Hyderabad, India during October 21-23, 2024.



Meghraj Goswami

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DATE OF BIRTH: 8 MARCH 2005
RESUME AS ON: 19 FEBRUARY 2026

EDUCATION

Institute: [Birla Institute of Technology and Science \(BITS\) Pilani](#) – Hyderabad Campus, **INDIA**

Programs of study being pursued (dual major):
B.E. Civil Engineering, M.Sc. (Integrated) Mathematics

Status: Currently in 4th year. Graduating in May 2027.

CGPA: 8.46/10 (Current)

Relevant Course works:

- **Engineering:** Fluid Mechanics, Hydraulic Engineering, Hydrology, Solid Mechanics, Structural Analysis, Steel & Concrete Structures, Soil Mechanics, Environmental Engineering, Surveying, Transportation Engineering, Building Materials.
- **Mathematics:** Numerical Analysis, Optimization, ODE, PDE, Linear Algebra, Probability & Statistics, Statistical Inference, Tensor Calculus, Functional Analysis, Graphs & Networks, Real & Complex Analysis, Cryptography, Differential Geometry.
- **Computing:** Computer Programming, Applications of AI in Civil Engineering, Discrete Mathematics, Scientific Computing.

TECHNICAL PROFICIENCY

SCIENTIFIC COMPUTING

- **Languages:** Python (Primary), C/C++, MATLAB/GNU Octave, R (Statistical Analysis), JavaScript/TypeScript, Java, Julia.
- **Libraries:** NumPy, SciPy, Pandas, Matplotlib/Seaborn, Scikit-learn
- **Mathematics:** Numerical stability analysis, Green's functions, Finite Difference/Element workflows, Lattice Boltzmann Method, Lattice-based Post Quantum Cryptography.
- **Data Processing:** Large-scale climate data analysis (NetCDF/HDF5), GIS-based analysis.

ENGINEERING SOFTWARE & TOOLS

- **CFD & Simulation:** OpenFOAM, STAAD.Pro, ETABS, WaterGEMS
- **CAD & GIS:** AutoCAD, Revit, QGIS, Google Earth Engine
- **DevOps & Platforms:** Linux, Git, Docker, LaTeX, SQL & MongoDB, Familiarity with High-Performance Computing (HPC)
- **Web & Mobile Full-Stack:** Flask/Django (Python), R Shiny, Dart (Flutter), Kotlin (Jetpack Compose), NodeJS, SvelteKit

RESEARCH & ACADEMIC PROJECTS

CONTINUUM & COMPUTATIONAL MECHANICS

1. **Green's Function for Elastic Wave Equation** – [Reports](#)

Advisor: Dr. [Sumit Kumar Vishwakarma](#), Dept. of Mathematics
Simulated and derived analytical solutions for wave propagation in elastic media with representation via Green's function, followed by analysis of Surface & Interfacial Waves.

2. **Lattice Boltzmann Method** (Ongoing project)

Advisor: Dr. [Jagadeesh Anmala](#), Dept. of Civil Engineering Contd. ...

LEADERSHIP & OUTREACH

Music Head | Swaranjali
(BITS Pilani - Hyderabad)

- Led the university's *Classical Music & Dance Club*. Organized campus-wide cultural events, managed team logistics, and conducted workshops.
- Developed and deployed the official club website, [Swaranjali Web](#).

Workshop Instructor

- [NTRU Cryptosystem – Lattice-based Post Quantum Cryptography](#)
- [Introduction to Web Development](#)
- [Introduction to Peer-to-Peer Networking](#)
- [Dart/Flutter Workshop at ACM BPHC](#)
- [Introduction to Classical Music](#)
- [Linux Dual-Booting Workshop](#)

Member | Clubs and Chapters
(BITS Pilani - Hyderabad)

- *Swaranjali*: Indian Music and Classical Dance club.
- CRUX: Programming and Computing club.
- CEA: Civil Engineering Association.
- *Music Club*: Contemporary and Western music club.
- BITS-SOS: Society for Open Software.
- ACM BPHC: Student Chapter for Association for Computing Machinery.
- GDSC: Google Developer Student Club.

LANGUAGES KNOWN & FLUENCY

English (International)

Speaking Writing Reading
●●●●● | ●●●●● | ●●●●●

Hindi (an Official language of India)

Speaking Writing Reading
●●●●● | ●●●●● | ●●●●●

Bengali (a Classical Indian Language)

Speaking Writing Reading
●●●●● | ●●●●○ | ●●●●○

French

Speaking Writing Reading
●●●○○ | ●●●●○ | ●●●●○

INTERNATIONAL EXPERIENCE

Travelled in the *US, Ireland, UK, France, Switzerland* and *Italy*, besides traveling throughout the length and breadth of *India*.

GEOSPATIAL & REMOTE SENSING ANALYSIS

1. PS-I Project on Antarctic Sea Ice Dynamics – [GitHub Link](#)

Advisor: Dr. [Alvarinho J. Luis](#), Scientist - F, NCPOR
Large scale data processing of climatic and cryospheric datasets (ERA5, NSIDC Index), analysis of polar ice dynamics.

2. Land Use/Land Cover Change Detection – [Link 1](#), [Link 2](#)

Advisor: Dr. [Abhradeep Majumder](#), Dept. of Civil Engineering
Quantification of LULC changes using DynamicWorld dataset for environmental monitoring and urban planning applications.

HYDROLOGY & MACHINE LEARNING

1. HydrologyFFA: Flood Frequency Analysis – [R Shiny Application](#)

Advisors: Dr. [K. Srinivasa Raju](#), Dr. [Vasan Arunachalam](#)
Interactive tool for Flood Frequency Analysis and Extreme Value Statistics for hydrological risk assessment. Deployed as a web application for accessibility to practitioners.

2. Sea-Ice Concentration Prediction (ML) – [GitHub Link](#)

Multi-Layer Perceptron (MLP) framework and GUI tool to predict sea-ice concentration using meteorological reanalysis data, optimizing hyperparameters for predictive accuracy.

EXTRACURRICULAR SKILLS

CLASSICAL MUSIC & ARTS

- **Indian Classical Music**: Sarod player with 10+ years of training and performance under the tutelage of [Pt. Mukesh Sharma](#).
- **Western Music**: Basic level, having experience in Keyboard.

TECHNICAL PROJECTS

- **NaadGen**: Web app to document Classical Music compositions
- **DailyRaga**: Mobile app to refer to attributes of Classical Ragas
- **Chronofactorem** (team): Web app to draft semester timetables
- **PyWaMG** (deprecated): Python package for chat automation

PROFESSIONAL EXPERIENCE & CERTIFICATIONS

1. Summer Training on Remote Sensing & GIS, India Space

Academy (ISA) | Jul 2025 – Aug 2025 | [Certificate](#)
Completed an intensive training program focused on fundamentals and applications of Remote Sensing & GIS tech.

2. National Centre for Polar and Ocean Research (NCPOR), Ministry of Earth Sciences | Research Intern, 2024 | [Certificate](#)

Conducted quantitative data analysis in the field of Earth Sciences to model climatological trends.

3. Smart India Hackathon | Finalist, 2024 | [Slides](#), [Demo](#)

Collaborated on rapid prototyping of engineering systems for a premier nationwide initiative by the Government of India to solve pressing real-world challenges.

4. Open Source, Freelance & Technical Community

Contributed actively in developer communities (*IRC/Discord/Online Forums*). Assisted peers with software implementations, documentation and debugging.

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